Anthony Gruber, Ph.D.

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Albuquerque, NM 87111

Education

Ph.D., Mathematics, Texas Tech University, Lubbock, TX — 2019

M.S., Mathematics, Texas Tech University, Lubbock, TX — 2017

B.G.S., Music Performance/Chemistry/Mathematics, Texas Tech University, Lubbock, TX — 2015

Professional Experience

Sandia National Laboratories — Albuquerque, NM
Senior Member of Technical Staff — *Sep 2024–Present*

- Conducting funded research in applied mathematics with emphasis on structure-preserving and variational approaches.
- Regularly publishing findings in high-impact journals, presenting at international conferences, mentoring graduate students and postdocs.

John von Neumann Fellow — *Sep 2022–Sep 2024*

- Led self-directed research funded by DOE ASCR and Sandia LDRD programs.
- Developed nonintrusive and variationally consistent methods for surrogate modeling and graph information processing.
- Collaborated with and advised by Pavel Bochev, Nat Trask, and Irina Tezaur.

Florida State University — Tallahassee, FL and Columbia, SC
Postdoctoral Research Associate — *Jan 2021–Aug 2022*

- Designed algorithms for function approximation and reduced-order modeling related to ocean dynamics simulations.
- Collaborated with and advised by Prof. Max Gunzburger (FSU), Prof. Lili Ju, and Prof. Zhu Wang (University of South Carolina).

Texas Tech University — Lubbock, TX and San José, Costa Rica
Assistant Professor of Practice — *Aug 2019–Dec 2020**

- Directed the mathematics program at TTU's satellite campus in Costa Rica.
- Taught a 2-2 load of mathematics courses, conducted research, and coordinated with TTU faculty and administration state-side to further the University mission in Costa Rica.
- *Remained employed on unpaid leave until Aug. 2022.

Graduate Part-Time Instructor *Aug 2015–Aug 2019*

- Served as instructor of record for a 2-2 load of mathematics courses annually.
- Responsible for all aspects of instruction, including writing/delivering lectures, assigning homework, and writing/grading exams.
- Funded through scholarships and endowments at TTU.

Oak Ridge National Laboratory Oak Ridge, TN
NSF Graduate Research Fellow *June 2018–Aug 2018*

- Developed "active manifolds" (see publications below) under Dr. Robert Bridges, applying geometric methods to high-dimensional function approximation problems.
- Produced mathematical and computational results specially selected for presentation to the leaders of the Computing and Computational Sciences Division at ORNL.
- Funded through the NSF Mathematical Sciences Graduate Internship (NSF MSGI) program.

University of Texas at Dallas Richardson, TX
NSF Research Intern *May 2014–Aug 2014*

- Worked under Prof. Manuel Quevedo to design, construct, and characterize TiSi and CrB2-Si-SiC thin-film resistors (TFRs) using a combination of lithography, x-ray photoelectron spectrometry, and Hall-effect measurements.
- Improved resistivity of previous thin-film resistors by 30% through optimized Ti:Si ratios.
- Funded through the NSF Research Experiences for Undergraduates (NSF REU) program.

Research Awards and Funding

Team member, "Applied mathematics and Scientific Computing Ecosystem for the New Digital era (ASCEND)", DoE ASCR. 2025-2030

Team member, "Data-Driven Models Accelerating Cavity Electromagnetic Response Computations (DD-MAkER)", Sandia LDRD award. 2024-2027

Team member, "Accelerating the Analyst Workflow: Adaptive Hybrid Models for Domain Decomposition (AHeaD)", Sandia LDRD award. 2024-2027

Team member, "Reduced-Order Model-Enabled Multifidelity Uncertainty Quantification, NNSA ASC L2 Milestone. 2024-2025

PI, "Learning Operators for Structure-Informed Surrogate Models", Sandia LDRD award. 2022-2024

Team member, "Multifaceted Mathematics for Predictive Digital Twins (M2dt)", DoE ASCR MMICC. 2022-2027

Team member, "Scalable, Efficient and Accelerated Causal Reasoning Operators Graphs and Spikes for Earth and Embedded Systems (SEA-CROGS)", DoE ASCR MMICC. 2022-2027

Awardee, John von Neumann Fellowship, Sandia National Laboratories 2022-2024

Postdoc, "Efficient and Scalable Time-Stepping Algorithms and Reduced-Order Modeling for Ocean System Simulations", DoE grant DE-SC0020418. 2021-2022

Awardee, NSF Mathematical Sciences Graduate Internship 2018

Awardee, NSF REU Internship 2014

Publications

Journal Articles

19. I. Tezaur, E. Parish, A. Gruber, I. Moore, C. Wentland, and A. Mota, "Hybrid coupling with operator inference and the overlapping Schwarz alternating method," *International Journal for Numerical Methods in Engineering*, (to appear).
18. A. Vijaywargiya, S. A. McQuarrie, and A. Gruber, "Tensor parametric Hamiltonian operator inference," *SIAM Journal on Applied Dynamical Systems*, (to appear).
17. X. He, Y. Shin, A. Gruber, S. Jung, K. Lee, and Y. Choi, "Thermodynamically consistent latent dynamics identification for parametric systems," *Transactions on Machine Learning Research (TMLR)*, 2026.
16. A. Gruber and I. Tezaur, "Variationally consistent Hamiltonian model reduction," *SIAM Journal on Applied Dynamical Systems*, pp. 376–414, 2025.
15. A. Gruber and I. Tezaur, "Canonical and noncanonical Hamiltonian operator inference," *Computer Methods in Applied Mechanics and Engineering*, vol. 416, p. 116334, 2023.
14. A. Gruber, Á. Pámpano, and M. Toda, "Instability of closed p-elastic curves in $\mathbb{S}^2$," *Analysis and Applications*, pp. 1–27, 2023.
13. A. Gruber, "Parallel Codazzi tensors with submanifold applications," *Mathematische Nachrichten*, vol. 00, pp. 1–11, 2023.
12. A. Gruber, M. Gunzburger, L. Ju, R. Lan, and Z. Wang, "Multifidelity Monte Carlo estimation for efficient uncertainty quantification in climate-related modeling," *Geoscientific Model Development*, vol. 16, no. 4, pp. 1213–1229, 2023.
11. A. Gruber, M. Gunzburger, L. Ju, and Z. Wang, "Energetically consistent model reduction for metriplectic systems," *Computer Methods in Applied Mechanics and Engineering*, vol. 404, p. 115709, 2023.
10. Y. Teng, Z. Wang, L. Ju, A. Gruber, and G. Zhang, "Level set learning with pseudoreversible neural networks for nonlinear dimension reduction in function approximation," *SIAM Journal on Scientific Computing*, vol. 45, no. 3, pp. A1148–A1171, 2023.
9. A. Gruber, Á. Pámpano, and M. Toda, "On p-Willmore disks with boundary energies," *Differential Geometry and its Applications*, vol. 86, p. 101971, 2023.
8. A. Gruber, M. Gunzburger, L. Ju, and Z. Wang, "A multifidelity Monte Carlo method for realistic computational budgets," *Journal of Scientific Computing*, vol. 94, no. 1, 2022.
7. A. Gruber, M. Toda, and H. Tran, "Stationary surfaces with boundaries," *Annals of Global Analysis and Geometry*, vol. 62, no. 2, pp. 305–328, 2022.
6. A. Gruber, M. Gunzburger, L. Ju, and Z. Wang, "A comparison of neural network architectures for data-driven reduced-order modeling," *Computer Methods in Applied Mechanics and Engineering*, vol. 393, p. 114764, 2022.

5. A. Gruber, “Planar immersions with prescribed curl and Jacobian determinant are unique,” *Bulletin of the Australian Mathematical Society*, vol. 106, no. 1, pp. 126–131, 2022.
4. A. Gruber, M. Gunzburger, L. Ju, Y. Teng, and Z. Wang, “Nonlinear level set learning for function approximation on sparse data with applications to parametric differential equations,” *Numerical Mathematics: Theory, Methods and Applications*, vol. 14, no. 4, pp. 839–861, 2021.
3. A. Gruber, Á. Pámpano, and M. Toda, “Regarding the Euler–Plateau problem with elastic modulus,” *Annali di Matematica Pura ed Applicata*, vol. 200, no. 5, pp. 2263–2283, 2021.
2. A. Gruber and E. Aulisa, “Computational p-Willmore flow with conformal penalty,” *ACM Transactions on Graphics (TOG)*, vol. 39, aug 2020.
1. A. Gruber, M. Toda, and H. Tran, “On the variation of curvature functionals in a space form with application to a generalized Willmore energy,” *Annals of Global Analysis and Geometry*, vol. 56, no. 1, pp. 147–165, 2019.

Articles in Refereed Conference Proceedings

9. C. Jing, U. B. Mudiyanselage, W. Cho, M. Jo, A. Gruber, and K. Lee, “Meta-learning structure-preserving dynamics,” in *Forty-third International Conference on Machine Learning (ICML)*, 2026.
8. A. Gruber, K. Lee, H. Lim, N. Park, and N. Trask, “Efficiently parameterized neural metriplectic systems,” in *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
7. E. Aulisa, A. Gruber, and M. Toda, “Generalized Willmore energies and applications,” in *Geometry, Integrability and Quantization, Papers and Lecture Series*, vol. 29, pp. 1–10, 2024.
6. M. Kang, D. Lee, W. Cho, K. Lee, A. Gruber, N. Trask, Y. Hong, and N. Park, “Can we pre-train ICL-based SFMs for the zero-shot inference of the 1D CDR problem with noisy data?,” in *NeurIPS Workshop on Foundation Models for Science: Progress, Opportunities, and Challenges*, 2024.
5. A. Gruber, K. Lee, and N. Trask, “Reversible and irreversible bracket-based dynamics for deep graph neural networks,” in *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS)*, 2023.
4. A. Gruber and E. Aulisa, “Quaternionic remeshing during surface evolution,” *AIP Conference Proceedings*, vol. 2425, no. 1, p. 330003, 2022.
3. A. Gruber, M. Toda, and H. Tran, “Willmore-stable minimal surfaces,” *AIP Conference Proceedings*, vol. 2425, no. 1, p. 330004, 2022.
2. E. Aulisa, A. Gruber, M. Toda, and H. Tran, “New developments on the p-Willmore energy of surfaces,” in *Proceedings of the Twenty-First International Conference on Geometry, Integrability and Quantization*, vol. 21, pp. 57–66, 2020.
1. R. Bridges, A. Gruber, C. Felder, M. Verma, and C. Hoff, “Active manifolds: A non-linear analogue to active subspaces,” in *Proceedings of the 36th International Conference on Machine Learning (ICML)*, vol. 97, pp. 764–772, 2019.

Submitted Articles

11. I. Moore, P.-H. Tsai, A. Gruber, I. Farcas, C. Wentland, I. Tezaur, and T. Iliescu, "Learning long-term stable operator inference reduced-order models of fluid flows through online spatial filtering," (under review).
10. R. Roy-Chowdhury, M. S. Nabizadeh, O. Gross, A. Gruber, and A. Chern, "Vakonomic fluids," (under review).
9. A. Gruber, R. Roy-Chowdhury, I. Tezaur, and N. Urban, "Flexible and stable dynamics discovery with Onsager's variational principle," (under review).
8. A. Vijaywargia, E. C. Cyr, and A. Gruber, "Structure-aware tensorial model reduction," (under review).
7. E. Parish, A. Gruber, P. Blonigan, and I. Tezaur, "NN-OpInf: an operator inference approach using structure-preserving composable neural networks," (under review).
6. A. M. Propp, M. Perego, E. C. Cyr, A. Gruber, A. Howard, A. Heinlein, P. Stinis, and D. Tartakovsky, "Domain-decomposed graph neural network surrogate modeling for ice sheets," (under review).
5. J. A. Actor, A. Gruber, and E. C. Cyr, "Deriving transformer architectures from implicit multinomial regression," (under review).
4. S. Hong, F. Wu, J. Kim, A. Gruber, N. Park, and K. Lee, "Rethinking privacy in neural networks through continuous-time dynamics," (under review).
3. J. Park, M. Kang, D. Lee, W. Cho, K. Lee, A. Gruber, Y. Hong, and N. Park, "PDE-PFN: Prior-data fitted neural PDE solver," (under review).
2. J. A. Actor, A. Gruber, E. C. Cyr, and N. Trask, "Gaussian variational schemes on bounded and unbounded domains," (under review).
1. A. Gruber and E. Aulisa, "Quasiconformal mappings with surface domains," (under review).

Feature Articles and News Releases

2. Distinguished fellowships offer research opportunities to some of the nation's best and brightest. Sandia National Laboratories 2022-2023 Academic Programs Collaboration Report, pp. 119.
1. Meet a Participant: Anthony Gruber. NSF Mathematical Sciences Graduate Internship, *Oak Ridge Institute for Science and Education (ORISE)*, 2019.

Other

9. A. Gruber, G. Geraci, P. J. Blonigan, E. Parish, and M. Eldred, "Advancements in reduced order model-based multifidelity uncertainty quantification for high-speed flows," AIAA SciTech Forum, American Institute of Aeronautics and Astronautics, January 2026.
8. R. Roy-Chowdhury, I. Tezaur, N. Urban, and A. Gruber, "Onsager's variational principle for solving inverse problems," *CSRI Summer Proceedings*, 2025.
7. T. T. Burnett, A. Gruber, and E. C. Cyr, "Towards structure-preserving reduced-order modeling of particle-in-cell systems," *CSRI Summer Proceedings*, 2025.

6. A. M. Propp, M. Perego, E. C. Cyr, A. Gruber, A. Howard, A. Heinlein, P. Stinis, and D. Tartakovsky, "Scalable ice sheet surrogate modeling via domain-decomposed, physics-inspired graph neural networks," *CSRI Summer Proceedings*, 2025.
5. C. Rodriguez, I. Tezaur, A. Mota, A. Gruber, E. Parish, and C. Wentland, "Transmission conditions for the non-overlapping schwarz coupling of full order and operator inference models," *CSRI Summer Proceedings*, 2025.
4. A. Gruber, "Learning operators for structure-informed surrogate models," tech. rep., Sandia National Lab.(SNL-NM), Albuquerque, NM (United States), 2024.
3. A. Vijaywargia, S. A. McQuarrie, and A. Gruber, "Tensor parametric operator inference with structure information," *CSRI Summer Proceedings*, 2024.
2. I. R. Moore, C. Wentland, A. Gruber, and I. Tezaur, "Domain decomposition-based coupling of operator inference reduced order models via the Schwarz alternating method," *CSRI Summer Proceedings*, 2024.
1. A. Gruber, *Curvature Functionals and p-Willmore Energy*. PhD thesis, Texas Tech University, 2019.

Organization

Conferences and Workshops

1. "Geometric and Topological Structure-Preserving Scientific Machine Learning", ICERM workshop, June 7-11, 2027.

Conference Minisymposia

11. "Recent developments in structure-preserving data-driven methods for fluids", *Minisymposium*, ICOSAHOM, July 5-9, 2027.
10. "Structure-Preserving Data-Driven Models for Predictive Digital Twins", *Minisymposium*, IACM Digital Twins in Engineering, March 8-12, 2027.
9. "Structure-preserving Uncertainty Quantification", *Minisymposium*, SIAM Conference on Uncertainty Quantification, March 22-25, 2026.
8. "Geometric Mechanics Formulations and Structure-Preserving Discretizations for Models of Physical Systems", *Minisymposium*, 18th U.S. National Congress on Computational Mechanics, July 20–24, 2025.
7. "Intersections of Scientific Machine Learning and Classical Computational Approaches", *Minisymposium*, 18th U.S. National Congress on Computational Mechanics, July 20–24, 2025.
6. "Structure-Preserving Model Reduction for Large-Scale Systems", *Minisymposium*, SIAM Computational Science and Engineering, March 3–7, 2025.
5. "Geometric Mechanics Formulations and Structure-Preserving Discretizations for Models of Physical Systems", *Minisymposium*, SIAM Joint Mathematics Meetings, January 8–11, 2025.
4. "Advances in Machine Learning on Graphs for Physical Sciences and Data Analysis", *Minisymposium*, SIAM Mathematics of Data Science, October 21–25, 2024.

3. "Geometric Mechanics Formulations and Structure-Preserving Discretizations for Continuum Mechanics and Kinetic Models", *Minisymposium*, 16th World Congress on Computational Mechanics, July 21–26, 2024.
2. "Geometric Mechanics Formulations and Structure-Preserving Discretizations for Continuum Mechanics", *Minisymposium*, 17th U.S. National Congress on Computational Mechanics, July 26–30, 2023.
1. "Elastic Curves and Surfaces with Applications and Numerical Representations", *Special Session #54*, 18th International Conference of Numerical Analysis and Applied Mathematics, September 17–23, 2020.

Research Visits Hosted or Cohosted

- Oliver Gross, Sandia National Laboratories. April 7-9, 2026.
- Shashikanth Banavara, Sandia National Laboratories. June 16-19, 2025.
- Ionut Farcas, Sandia National Laboratories. Nov. 18-22, 2024.
- Artur Palha, Sandia National Laboratories. July 28, 2023.
- Marc Gerritsma, Sandia National Laboratories. July 28, 2023.

Leadership, Mentoring, and Professional Service

Formal Mentoring: Served as formally designated mentor/advisor with documented role.

- James Cash (6/2026-Present), M2dt summer graduate intern, U Alberta.
  Project: Variational integrators for viscoelastic material models.
  * Served as primary mentor.
- Trishit Mondal (6/2026-Present), Sandia summer graduate intern, WPI.
  Project: Reinforcement learning for accelerating the Schwarz Alternating Method.
  * Served as supporting mentor.
- Ritoban Roy-Chowdhury (6/2025–7/2026), M2dt year-round undergraduate intern, UC San Diego.
  Project: Learning phase-field models with Onsager's variational principle.
  * Served as primary mentor.
- Tage Burnett (5/2025–8/2025), SEA-CROGS summer graduate intern, Georgia Tech.
  Project: Structure-preserving ROMs for PIC methods.
  * Served as primary mentor.
- Sohyeon Jung (8/2024–Present), Ph.D. student with Kookjin Lee at Arizona State University.
  Project: Model reduction with neural metriplectic systems.
  * Served on Ph.D. committee.
- Arjun Vijaywargia (6/2024–5/2025), M2dt year-round graduate intern, U Notre Dame.
  Project: Tensor parametric Hamiltonian operator inference.
  * Served as primary mentor.
- Ian Moore (6/2024–7/2026), M2dt year-round graduate intern, Virginia Tech.
  Project: Schwarz coupling of operator inference models for convection-dominated physics.
  * Served as supporting mentor.

- Madusha Atampalage (2019–2021), Ph.D. student with Magdalena Toda at Texas Tech.
  Project: Topics in Minimal Surfaces and Applications.
  * Served on Ph.D. committee.

Additional Mentoring: Contributed substantively to mentee's professional development without formal documentation.

- Mohammad (Sina) Nabizadeh (3/2025–Present), Postdoc with Justin Solomon, MIT.
  Project: Fluid Implicit Particles on Coadjoint Orbits.
- Oliver Gross (1/2026–Present), Postdoc with Albert Chern, UC San Diego.
  Project: Vakonomic Fluid Simulation.
- Adrienne Propp (3/2024–1/2026), SEA-CROGS graduate intern, Stanford University.
  Project: Bracket-based graph neural networks for modeling ice sheets.
- Rishi Pawar (Summers 2023–2025), CSRI summer graduate intern, University of Arizona.
  Project: Interfacial flux prediction using neural ODEs and operator inference.
- Roxana Pohlmann (9/2023–1/2024), Visiting graduate student intern, TU Wien.
  Project: Reduced-order models to accelerate injection molding.
- Edward Huynh (5/2023–8/2023), CSRI summer graduate intern, UT Austin.
  Project: Obtaining inverse Sobolev-type inequalities for tanh neural networks.

Editorial and Review Activities

- Reviewer for journals including: *SIAM Journal of* {*Scientific Computing, Uncertainty Quantification, Applied Mathematics*}, *Computer Methods in Applied Mechanics and Engineering*, *Journal of Computational Physics*, *Journal of Scientific Computing*, *Physical Review* {*A, E*}, *Physica D*, *Journal of Geometry and Physics*, *Journal für die reine und angewandte Mathematik (Crelle's Journal)*, *Electronic Journal of Statistics*, *ACM Transactions on AI for Science*, *Technometrics*, *Expert Systems with Applications*, *Journal of Machine Learning for Modeling and Computing*, *Neural Networks*, *Geoscientific Model Development*, *IEEE Computing in Science and Engineering*, *Mathematics and Computers in Simulation*.
- Reviewer for conferences including: *NeurIPS*, *ICML*, *ICLR*, *AAAI*, *AISTATS*.
- Reviewer for funding organizations including: DoE Office of Science, Air Force Research Laboratory.
- Reviewer for lab-directed research and development (LDRD) awards: Sandia National Laboratories, Los Alamos National Laboratory.
- Reviewer for Sandia CSRI Summer Proceedings since 2023.

Panels and Outreach

- Panel Member, AI/ML Educational, 3rd annual SMART Exchange. March 5, 2026.
- Panel Member, Alumni Panel (virtual), NSF-MSGI Virtual Symposium. August 23, 2023.
- Panel Member, Early Career Panel (virtual), Association of Women in Mathematics, Texas Tech University, Lubbock, TX. April 21, 2022.

Teaching

- *Advanced Calculus I* (400 level)
  - One section, Fall 2020, TTU-CR.
- *Foundations of Algebra I* (300 level)
  - One section, Fall 2020, TTU-CR.
- *Higher Math for Engineers and Scientists II* (300 level)
  - One section, Spring 2019, TTU-CR.
- *Higher Math for Engineers and Scientists I* (300 level)
  - Two sections, Spring 2017, TTU.
  - Large section, Spring 2018, TTU.
  - Virtual section, Fall 2018, TTU.
  - One section, Spring 2020, TTU-CR.
- *Intro to Critical Reasoning and Proof* (300 level)
  - One section, Fall 2019, TTU-CR.
- *Calculus III with Applications* (200 level)
  - Large section, Fall 2016, TTU.
  - Two sections, Fall 2017, TTU.
  - Virtual section, Summer 2019, TTU.
  - One section, Fall 2019, TTU-CR.
  - One section, Spring 2019, TTU.
  - One section, Fall 2020, TTU-CR.
- *College Algebra* (100 level)
  - One section, Summer 2016, TTU.
- *Intro to Data Analytics* (general knowledge)
  - Short course, Fall 2020, TTU-CR.

Professional Presentations

Invited Presentations (expenses/honorarium paid)

7. "Structure-preserving SciML with (multi-)Linear Parameterizations" *2026 NSF RTG Workshop on Mathematical Foundations for Scientific AI and Data Science*, Auburn University, Auburn, AL. (50 min; June 13, 2026)

6. "Variational Approaches to Surrogate Modeling and Information Processing" *Santa Fe Institute Workshop on Dynamical Systems and Graph Theory Approaches to Digital Twins*, Santa Fe, NM. (40 min; Aug. 13, 2025)

5. "Model Reduction for Accelerating Bracket-Based SciML" *IVADO Workshop on Accuracy and Efficiency in SciML*, Montréal, Canada. (30 min; June 26, 2025)

4. "Hamiltonian and Metriplectic Model Reduction" *Workshop on Geometric Mechanics Formulations for Continuum Mechanics*, Banff, Canada. (20 min; March 20, 2025)

3. "Data-Driven Dynamical Systems with Structural Guarantees" *Los Alamos National Labora-*

tory, Los Alamos, NM. (50 min; November 19, 2024)

2. "Calculus in Computer Graphics and Data Science" *Mathematics Seminar Series*, Cameron University, Lawton, OK. (50 min; October 19, 2021)

1. "Stationary Surfaces for Curvature Functionals" *63rd Texas Geometry and Topology Conference*, Texas Tech University, Lubbock, TX. (50 min; April 23, 2020)

Other Invited Presentations

11. "Variational Methods in Surrogate Modeling and Information Processing" *Computational and Applied Math / Data Science seminar* (virtual), Arizona State University, Tempe, AZ. (50 min; October 7, 2026)

10. "Recent Progress in Structure-Preserving Model Reduction and Hybrid Modeling" *S. Scott Collis Advanced Modeling & Simulations Seminar Series* (virtual), Rio Grande Consortium for Advanced Research on Exascale Simulation. (50 min; April 17, 2026)

9. "Bracket-Based Dynamics for Data-Driven Surrogate Modeling and SciML" *Center for Mathematics and Artificial Intelligence Colloquium* (virtual), George Mason University, Fairfax, VA. (50 min; March 28, 2025)

8. "Learning Metriplectic Systems and Other Bracket-Based Dynamics" *University of Vienna Mathematics Seminar* (virtual), Vienna, Austria. (50 min; June 19, 2024)

7. "Learning Bracket-Based Dynamical Systems for Property-Preserving Model Reduction" *Guest Lecture*, University of Pennsylvania, Philadelphia, PA. (50 min; April 23, 2024)

6. "Property-Preserving Model Reduction in Bracket-Based Dynamical Systems" *Applied Math Seminar Series*, University of New Mexico, Albuquerque, NM. (50 min; March 25, 2024)

5. "Data-Driven Dynamical Systems with Structural Guarantees" *S. Scott Collis Advanced Modeling & Simulations Seminar Series* (virtual), Rio Grande Consortium for Advanced Research on Exascale Simulation. (50 min; November 10, 2023)

4. "Data-Driven Dynamical Systems with Structural Guarantees" *Applied Mathematics and Machine Learning Seminar* (virtual), Texas Tech University, Lubbock, TX. (50 min; November 8, 2023)

3. "Property-Preserving Model Reduction for Conservative and Dissipative Systems" *Numerical Analysis of Galerkin ROMs Seminar Series* (virtual), INRIA Bordeaux, France. (50 min; October 10, 2023)

2. "Mathematics in Different Settings" *Hong Duc University Seminar* (virtual), Thanh Hóa, Vietnam. (30 min; May 20, 2023)

1. "Energetically Consistent Model Reduction for Hamiltonian and Metriplectic Systems" *CRUNCH Webinar* (virtual), Brown University, Providence, RI. (60 min; December 9, 2022)

Invited Minisymposium Talks

13. "Flexible and Stable Dynamics Discovery with Onsager's Variational Principle" *Minisymposium on Reduced-Order Modeling for Large-Scale and Challenging Applications*, SIAM TX/LA Sectional Meeting, Dallas, TX. (25 min; October 30, 2026)

12. "Multilinear Surrogates Modeling Wave-Type Phenomena" *Minisymposium on Advances in Model Order Reduction: Bridging Physics and Machine Learning*, WCCM-ECCOMAS 2026, Munich, Germany. (15 min; July 23, 2026)
11. "Goal-Oriented Model Construction for Multifidelity Approximate Control Variate Estimation" *Minisymposium on Recent Advances in Inverse Methods for Complex Systems under Uncertainty*, SIAM Conference on Uncertainty Quantification, Minneapolis, MN. (25 min; March 25, 2026)
10. "Tensor Parametric Hamiltonian Operator Inference" *Minisymposium on Recent Advances in Structure-Preserving Machine Learning and Reduced Order Modeling*, SIAM Conference on Computational Science and Engineering, Fort Worth, TX. (15 min; March 3, 2025)
9. "Structure-Informed Model Reduction of Maxwell-Type Systems" *Minisymposium on Discovering Evolution Equations Using Structure-Preserving Data-Driven Methods*, DTE & AICOMAS 2025, Paris, France. (15 min; February 19, 2025)
8. "Tensor Parametric Hamiltonian Operator Inference" *Minisymposium on Reduced Order Models for Convection-Dominated Flows: Modeling, Analysis, and Simulation*, Joint Mathematics Meetings, Seattle, WA. (30 min; January 10, 2025)
7. "Variationally Consistent Hamiltonian Model Reduction" *Minisymposium on Recent Advancements in Data-Driven Model Reduction: Theory, Algorithms, and Applications*, SIAM Mathematics of Data Science, Atlanta, GA. (25 min; October 21, 2024)
6. "Learning Metriplectic Systems and Other Bracket-Based Dynamics" *Minisymposium on Mathematics for Machine Learning*, Canadian Mathematical Society Summer Meeting, University of Saskatchewan, Saskatoon, Canada. (30 min; June 2, 2024)
5. "Data-Driven Surrogate Models for Bracket-Based Dynamical Systems" *Minisymposium on Data-Driven Methods for Circuits and Devices*, 2nd IACM Mechanistic Machine Learning and Digital Engineering for Computational Science Engineering and Technology, El Paso, TX. (20 min; September 27, 2023)
4. "Convolutional Neural Networks for Data Compression and Reduced-Order Modeling" *Minisymposium on Machine Learning for Large-Scale Scientific Data Analytics*, SIAM Mathematics of Data Science, San Diego, CA. (25 min; September 28, 2022)
3. "Computing Quasiconformal Mappings Between Immersed Surfaces" *AMS Fall Central Sectional*, University of Texas at El Paso, El Paso, TX. (20 min; September 17, 2022)
2. "Convolutional Neural Networks for Data Compression and Reduced Order Modeling" *SIAM SEAS Special Session on Deep Learning Methods for Data Driven Models*, Auburn University, Auburn, AL. (30 min; September 18, 2021)
1. "Codazzi Tensors with Parallel Mean Curvature" *AMS Special Session #1159: Geometry of Submanifolds and Integrable Systems* (virtual), University of Texas at El Paso, El Paso, TX. (25 min; September 12, 2020)

Contributed Presentations

12. "Enabling Flexible Simulation with Non-Intrusive Model Reduction" *Workshop on SciML*, University of Texas, Austin, TX. (Poster; September 25, 2025)

11. “Enabling Flexible Simulation with Non-Intrusive Model Reduction” *Workshop on Artificial Intelligence and Digital Twins for Earth Systems (AIDT4ES)*, University of Texas, Austin, TX. (Poster; September 22, 2025)

10. “Reversible and Irreversible Bracket-Based Dynamics for Deep Graph Neural Networks” *Minisymposium on Machine Learning on Graphs for Physical Sciences and Data Analysis*, SIAM Mathematics of Data Science, Atlanta, GA. (Poster; October 22, 2024)

9. “Property-Preserving Machine Learning of Metriplectic Systems” *Oden Institute Workshop on Scientific Machine Learning*, Austin, TX. (30 min; October 3, 2024)

8. “Flexible and Variationally Consistent Hamiltonian Model Reduction” *Model Reduction and Surrogate Modeling (MORe2024)*, La Jolla, CA. (30 min; September 10, 2024)

7. “Learning Metriplectic Systems from Full and Partial State Information” *Minisymposium on Geometric Mechanics Formulations and Structure-Preserving Discretizations for Continuum Mechanics and Kinetic Models*, 16th World Congress on Computational Mechanics, Vancouver, Canada. (20 min; July 23, 2024)

6. “Reversible and Irreversible Bracket-Based Dynamics for Deep Graph Neural Networks” *Advances in Neural Information Processing Systems*, New Orleans, LA. (Poster; December 12, 2023)

5. “Variational Consistency in Model Reduction for Conservative and Dissipative Systems” *Minisymposium on Data-Driven Methods—Solids, A Conference Celebrating the 80th Birthday of Thomas J.R. Hughes*, Austin, TX. (25 min; October 23, 2023)

4. “Canonical and Noncanonical Hamiltonian Operator Inference” *Minisymposium on Geometric Mechanics Formulations and Structure-Preserving Discretizations*, 17th U.S. National Congress on Computational Mechanics, Albuquerque, NM. (25 min; July 26, 2023)

3. “Canonical and Noncanonical Hamiltonian Model Reduction” *Workshop on Establishing Benchmarks for Data-Driven Modeling of Physical Systems*, University of Southern California, Los Angeles, CA. (30 min; April 6, 2023)

2. “Quaternionic Remeshing During Surface Evolution” *18th International Conference of Numerical Analysis and Applied Mathematics* (virtual), Rhodes, Greece. (30 min; September 17, 2020)

1. “Willmore-Stable Minimal Surfaces” *18th International Conference of Numerical Analysis and Applied Mathematics* (virtual), Rhodes, Greece. (30 min; September 17, 2020)

Invited Internal and Other Presentations

25. “Flexible Hybrid Modeling with Operator Inference and the Schwarz Alternating Method” *3rd Annual Sandia SMART Exchange*, Albuquerque, NM. (25 min; March 5, 2026)

24. “Neural Architectures and Surrogates with Guaranteed Dynamical Behavior” *Sandia Center 1400 Town Hall*, Albuquerque, NM. (10 min; April 10, 2025)

23. “Neural Architectures and Surrogates with Guaranteed Dynamical Behavior” *Sandia Computing and Information Sciences (CIS) Research Foundation External Review Board (ERB) Meeting*, Livermore, CA. (Poster; March 11, 2025)

22. "Learning on Graphs with Bracket-Based Dynamical Systems" *SEA-CROGS MMICC 3rd Year Review* (virtual). (15 min; November 14, 2024)
21. "Hamiltonian Structure-Preserving ROMs" *M2dt MMICC 3rd Year Review* (virtual). (15 min; November 6, 2024)
20. "Flexible and Variationally Consistent Hamiltonian Model Reduction" *Sandia Fellows Day*, Livermore, CA. (15 min; July 30, 2024)
19. "Cohomology-Aware Model Reduction" *Sandia M2dt MMICC Meeting* (virtual), Albuquerque, NM and Livermore, CA. (45 min; June 25, 2024)
18. "Data-Driven Dynamical Systems with Structural Guarantees" *Applications in Algebra Working Group*, Sandia National Laboratories, Albuquerque, NM. (50 min; March 7, 2024)
17. "Learning Operators for Structure-Informed Surrogate Models" *DOE ASCR PI Meeting*, Albuquerque, NM. (Poster; January 8, 2024)
16. "SNL Progress Highlights: Data-Driven Couplings (RT3.1) and Preservation of Geometric Structure in ROM (RT2.2)" *M2dt MMICC All-Hands Meeting*, Oden Institute, University of Texas at Austin, Austin, TX. (20 min; October 25, 2023, w/ Irina Tezaur)
15. "Data-Driven Surrogate Models for Bracket-Based Dynamical Systems" *Sandia SEA-CROGS MMICC Meeting*, Albuquerque, NM. (50 min; October 11, 2023)
14. "Tensor Methods for Metriplectic Systems" *Sandia M2dt MMICC Meeting* (virtual), Albuquerque, NM and Livermore, CA. (25 min; May 16, 2023)
13. "ROM Ideas for Hodge-de Rham Systems" *Sandia M2dt MMICC Meeting* (virtual), Albuquerque, NM and Livermore, CA. (25 min; December 6, 2022)
12. "Hamiltonian Operator Inference with Examples" *Sandia M2dt MMICC Meeting* (virtual), Albuquerque, NM and Livermore, CA. (25 min; November 1, 2022)
11. "Structure-Preserving ROM Ideas" *Sandia M2dt MMICC Meeting* (virtual), Albuquerque, NM and Livermore, CA. (25 min; October 18, 2022)
10. "Variationally Consistent Model Reduction" *Sandia Fellows Day*, Albuquerque, NM. (20 min; August 29, 2023)
9. "Artificial Neural Networks for Dimension Reduction and Reduced-Order Modeling" *Applied Mathematics Group*, Texas Tech University, Lubbock, TX. (50 min; September 30, 2021)
8. "Some Nonlinear PDEs in Computer Graphics and Data Science" *Mathematics Colloquium Series*, Texas Tech University, Lubbock, TX. (50 min; September 29, 2021)
7. "Optimal Quasiconformal Mappings with Prescribed Boundary" *Probability, Geometry, and Mathematical Physics Group* (virtual), Texas Tech University, Lubbock, TX. (50 min; April 7, 2021)
6. "Geometric Flows via Finite Element Methods" *Elasticity Group* (virtual), Texas Tech University, Lubbock, TX. (50 min; December 2, 2020)
5. "Variational Aspects of Curvature Functionals" *Elasticity Group*, Texas Tech University, Lubbock, TX. (50 min; September 2, 2020)

4. "Computing Stationary Solutions to p-Willmore Flow" *Applied Mathematics Group*, Texas Tech University, Lubbock, TX. (50 min; April 22, 2020)
3. "A Conformally-Adjusted Willmore Flow of Closed Surfaces" *Applied Mathematics Group*, Texas Tech University, Lubbock, TX. (50 min; May 8, 2019)
2. "Curvature Functionals and p-Willmore Energy" *Analysis Group*, Texas Tech University, Lubbock, TX. (50 min; April 29, 2019)
1. "Active Manifolds: A Geometric Approach to Dimension Reduction for Sensitivity Analysis" *Computational and Applied Mathematics Group*, Oak Ridge National Laboratory, Oak Ridge, TN. (50 min; August 1, 2018)